

FPGA Programming – VHDL Part

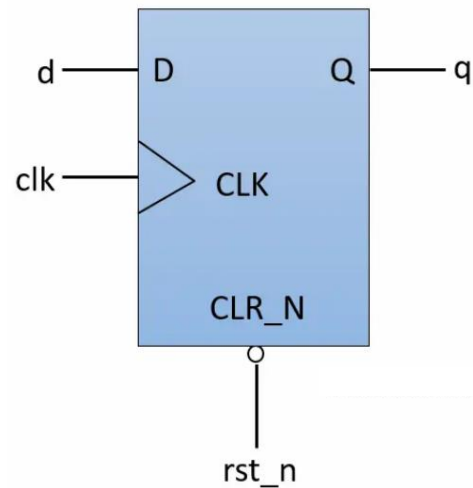
VHDL by Examples – Memory elements

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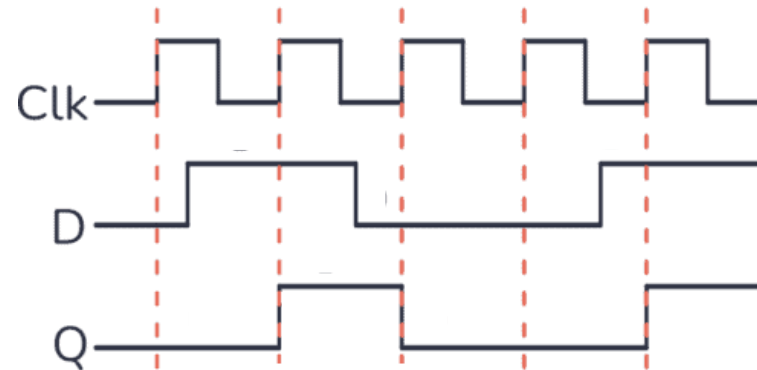
Memory elements

- Flip-flop D

$$Q^{n+1} = D^n$$



D^n	Q^{n+1}	$\overline{Q}^{n+1}$
0	0	1
1	1	0



Memory elements

- D Flip-flop in VHDL

```
entity RisingEdge_DFlipFlop is  
  port(  
    Q : out std_logic;  
    Clk :in std_logic;  
    D :in  std_logic  
  );  
end RisingEdge_DFlipFlop;
```

```
architecture Behavioral of RisingEdge_DFlipFlop is  
begin  
  process(Clk)  
  begin  
    if (rising_edge(Clk)) then  
      Q <= D;  
    end if;  
  end process;  
end Behavioral;
```

VHDL sequential

- Usually, all the assignments and instantiations in VHDL are concurrent
- Sequential statements can be used in sub-program (procedure and function) or **processes**

```
entity priority is
    port (a: in std_logic_vector(3 downto 0);
          y: out std_logic_vector(1 downto 0);
          valid: out std_logic);
end entity priority;

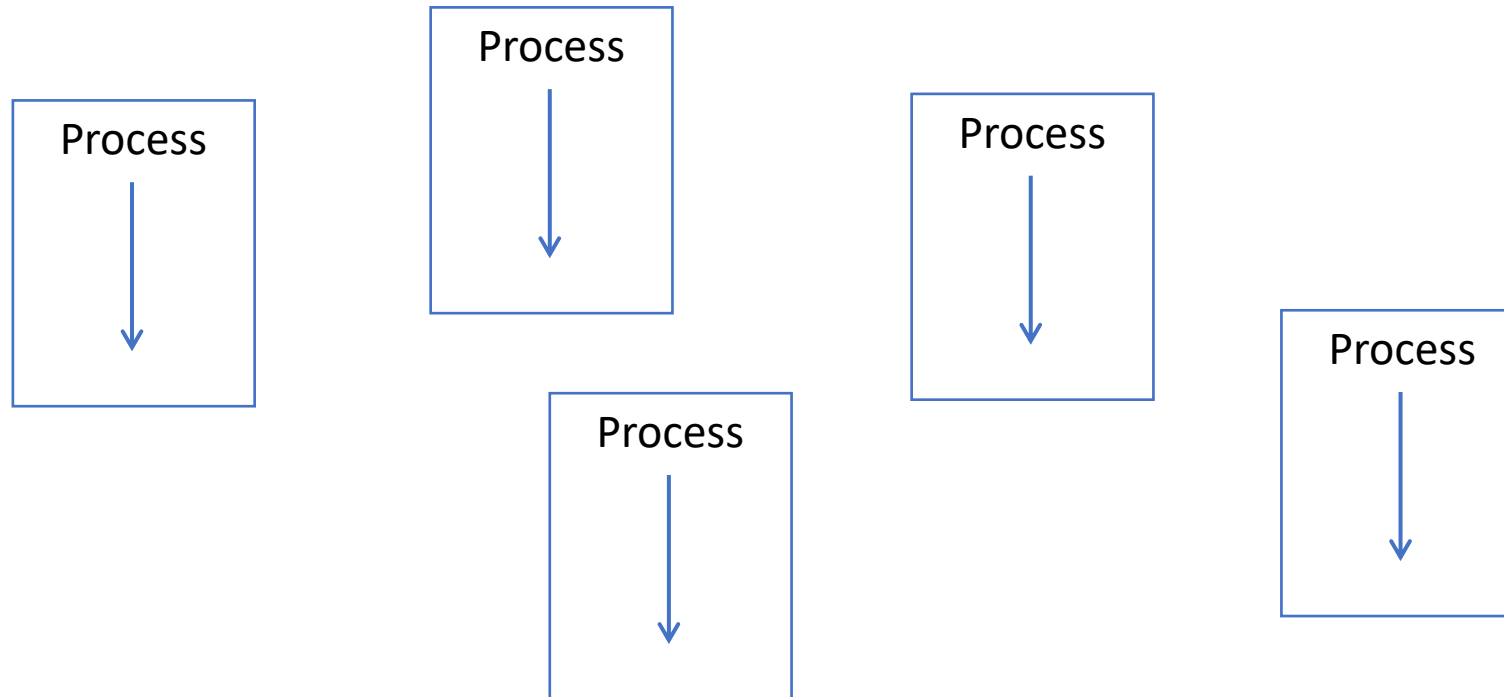
architecture Concurrent of priority is
begin
    y <= "11" when a(3) = '1' else
        "10" when a(2) = '1' else
        "01" when a(1) = '1' else
        "00" when a(0) = '1' else
        "00";
    valid <= '1' when a(0) = '1' or a(1) = '1'
        or a(2) = '1' or a(3) = '1' else
        '0';
end architecture Concurrent;
```

```
architecture Sequential of
priority is
begin
    process (a) is
    begin
        if a(3) = '1' then
            y <= "11";
            valid <= '1';
        elsif a(2) = '1' then
            y <= "10";
            valid <= '1';
        elsif a(1) = '1' then
            y <= "01";
            valid <= '1';
        elsif a(0) = '1' then
            y <= "00";
            valid <= '1';
        else
            y <= "00";
            valid <= '0';
        end if;
    end process;
end architecture Sequential;
```

```
architecture Sequential2 of
priority is
begin
    process (a) is
    begin
        valid <= '1';
        if a(3) = '1' then
            y <= "11";
        elsif a(2) = '1' then
            y <= "10";
        elsif a(1) = '1' then
            y <= "01";
        elsif a(0) = '1' then
            y <= "00";
        else
            y <= "00";
            valid <= '0';
        end if;
    end process;
end architecture Sequential2;
```

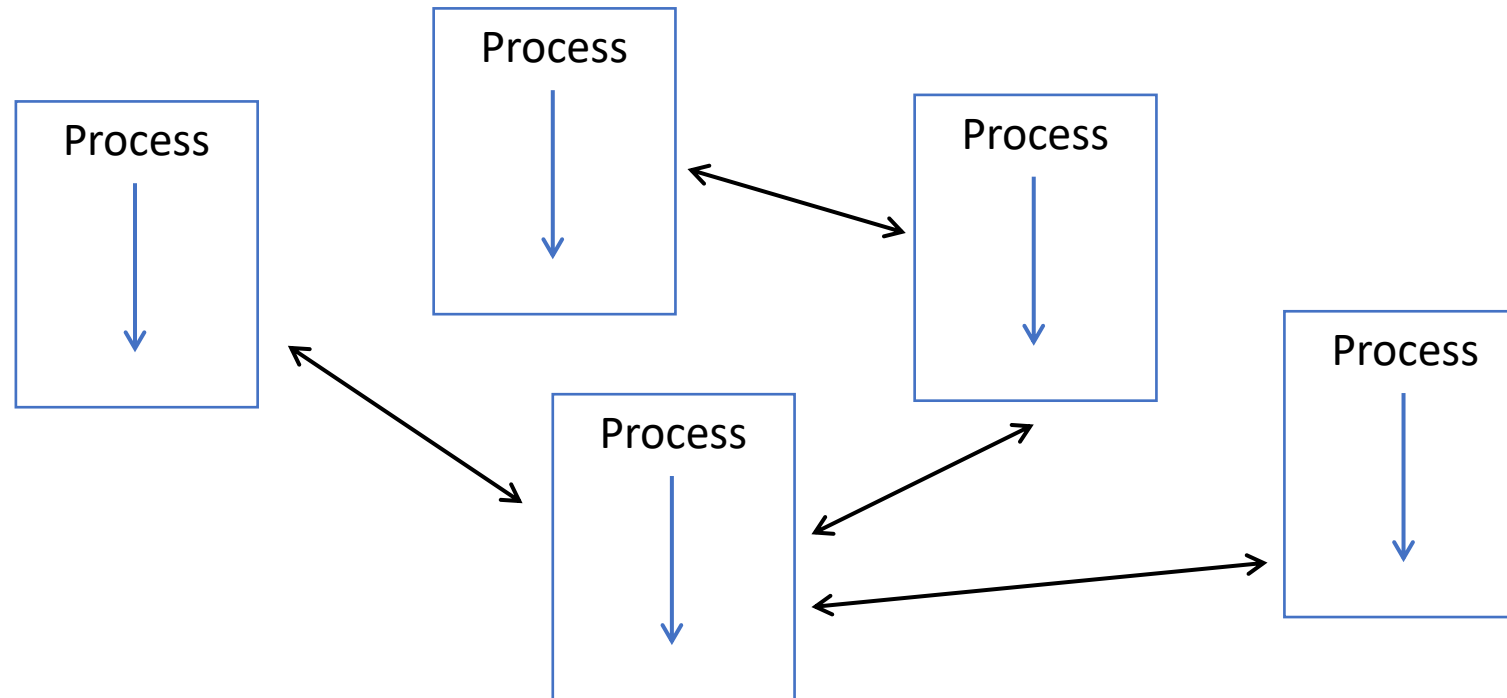
VHDL sequential

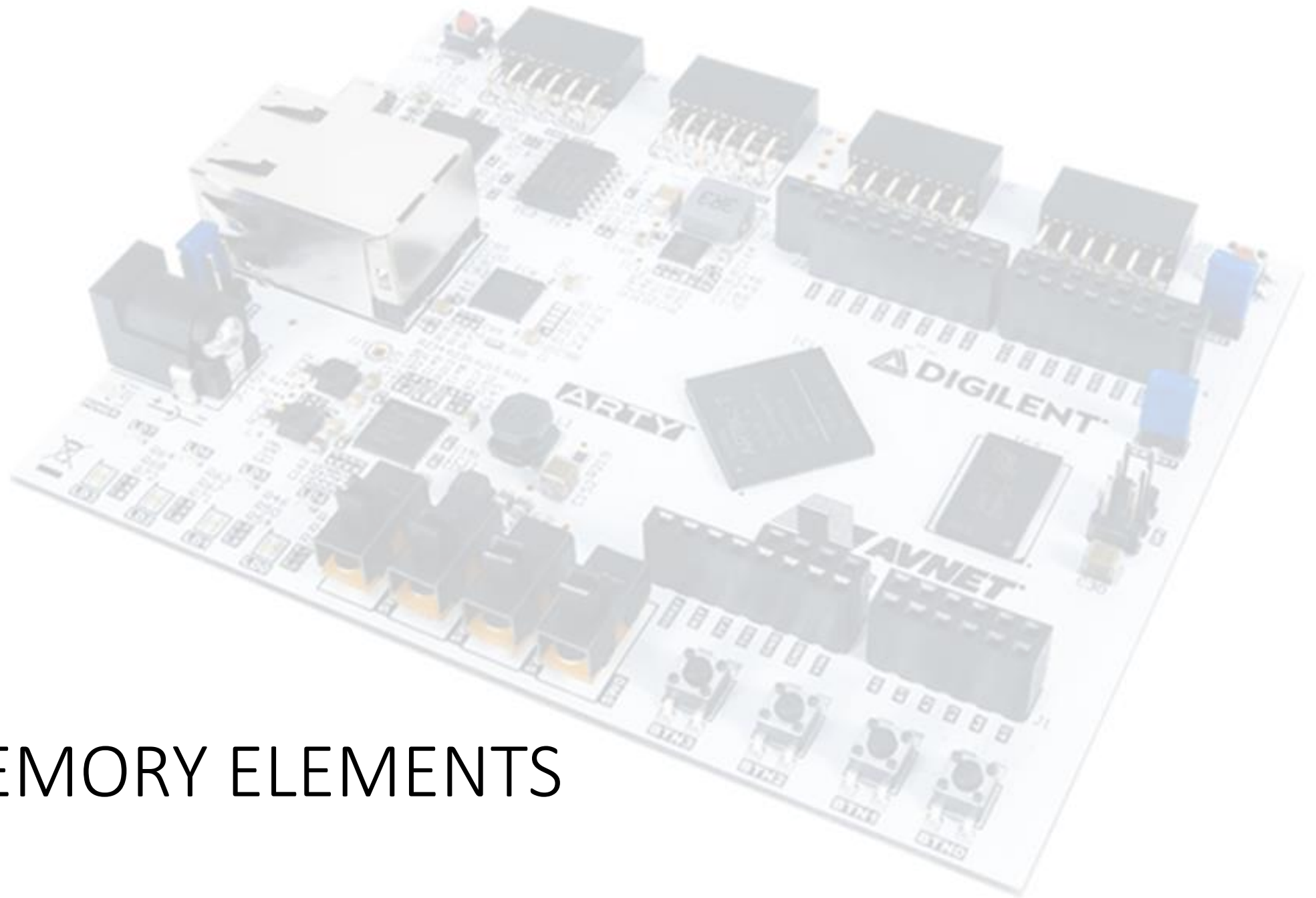
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VHDL sequential

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LAB MEMORY ELEMENTS